

A direct proof of the complete Crouzeix inequality

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September 26, 2026

Abstract

We give a direct proof of the sharp constant-two numerical-range inequality for matrix-valued polynomials in all finite base and coefficient dimensions. This resolves the complete Crouzeix conjecture in its matrix formulation, including matrices whose numerical ranges are points or line segments.

1 Introduction

The numerical range of a matrix $A \in M_n(\mathbb{C})$ is

$$W(A) = \{u^* A u : u \in \mathbb{C}^n, u^* u = 1\}.$$

It is a compact convex set containing the spectrum. For a normal matrix, the spectral theorem controls polynomial evaluation by values on the spectrum. For a nonnormal matrix, the spectrum alone cannot provide the same control: even a nonzero nilpotent matrix has spectrum $\{0\}$. The numerical range retains enough information to give a universal bound.

We allow matrix coefficients. If $P(z) = \sum_{k=0}^d B_k z^k$ with $B_k \in M_m(\mathbb{C})$, set

$$P[A] = \sum_{k=0}^d A^k \otimes B_k.$$

The base space $\mathbb{C}^n$ is the first tensor factor. All operator norms come from the Euclidean inner products and their Hilbert tensor products. The complete Crouzeix conjecture asks whether the constant two holds uniformly in the coefficient dimension m , the base dimension, and the degree.

Theorem 1.1. *For every pair of positive integers n, m , every $A \in M_n(\mathbb{C})$, every nonnegative integer d , and every polynomial $P(z) = \sum_{k=0}^d B_k z^k$ with $B_k \in M_m(\mathbb{C})$,*

$$\|P[A]\| \leq 2 \max_{z \in W(A)} \|P(z)\|. \quad (1.1)$$

The constant two cannot be decreased uniformly over n, m , and d .

The estimate includes point and segment numerical ranges. Its complete character is the uniformity in m : a scalar proof need not survive matrix amplification, because coefficient matrices need not commute. The argument below keeps their multiplication order, using the two products FG and GF separately.

Earlier bounds and related arguments

Crouzeix formulated the scalar constant-two question in 2004 and proved a universal complete bound of 11.08 in 2007 [7, 8]. Positive boundary representations developed by Delyon and Delyon [11] supply an important part of the numerical-range method. Crouzeix and Palencia combined a positive double-layer potential with the Cauchy transform of conjugate boundary data to obtain the complete bound $1 + \sqrt{2}$ [10]; Ransford and Schwenninger simplified its final norm argument [17]. For all base matrices of order two, Badea, Crouzeix, and Delyon obtained the complete constant two. In the nonnormal case, they make the conformal image of the matrix a contraction by a similarity with condition number at most two, using the numerical-range ellipse [3].

Recent scalar proofs include work of Jin [13], Lorist and Schwenninger [14], and Luo [15]. Lorist and Schwenninger's Theorem 3 applies to scalar rational functions of bounded Hilbert-space operators; their Section 3(iv) explains why the commutation used in that proof does not automatically persist under amplification. Åhag, Czyż, and Virtanen prove the complete inequality for base matrices of order at most three, with arbitrary coefficient dimension [1, Theorem 1.1]. The scalar results fix $m = 1$, while the latter complete result restricts the base dimension; Theorem 1.1 leaves these dimensions independent.

The use of Faber polynomials in numerical-range functional calculus has precedents in [4, 5]. Lorist and Schwenninger use a norm-attaining vector in their scalar argument [14, Lemma 1], while Luo chooses an explicit top singular pair [15, Section 3, equations (21)–(22)]. In their complete constant-two proof for weighted shift matrices, Crouzeix and Greenbaum use a scalar extremal function and a norm-attaining vector after a similarity reduction [9, Section 3, proof of Theorem 2]. Here a Faber coefficient estimate controls the two diagonal functionals obtained by placing a test polynomial on each side of the polynomial being estimated.

A separate structural proof of the complete inequality is given in [16]. For a bounded convex domain Ω with regular real-analytic boundary and $W(A) \subset \Omega$, it constructs a positive definite S with $\|S\|\|S^{-1}\| \leq 2$ such that $Sf(A)S^{-1}$ is a contraction, where $f : \Omega \rightarrow \mathbb{D}$ is an interior conformal map and $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. The same S , obtained from a minimizing metric, gives one continuous normalized positive boundary density representing the calculus of SAS^{-1} for every coefficient dimension and every matrix function holomorphic near $\overline{\Omega}$ [16, Theorems 1.2–1.3 and Proposition 2.1]. The direct proof below estimates one polynomial at a time. Its exterior kernel, marginal identities, and positive block comparison have counterparts in [16, Propositions 3.1 and 3.3 and Lemma 5.1]; Sections 2–3 prove the specific coefficient, resolvent, and block estimates used here.

How the coefficients give the sharp bound

First place $W(A)$ strictly inside an analytic convex domain Ω . An exterior conformal map h parametrizes its boundary by the unit circle. Expanding the Cauchy kernel at infinity defines scalar polynomials b_k of degree k . For $k \geq 1$, the boundary value $b_k \circ h$ has the positive Fourier term λ^k and an otherwise strictly negative expansion. In Section 2, a nonnegative kernel with integral one in each variable makes the map from the positive part to the negative part an L^2 contraction. Thus, for $G = \sum_k b_k C_k$, its squared boundary Hilbert–Schmidt norm lies between the square sum of its coefficient norms and that sum with weight two on every nonconstant coefficient.

For a polynomial F normalized by its boundary operator norm, choose a unit top singular pair x, y for $F[A]$. The resolvent $R(\lambda) = \lambda h'(\lambda)(h(\lambda)I - A)^{-1}$ supplies the coefficient arrays representing the three functionals $y^*G[A]x$, $x^*G[A]x$, and $y^*G[A]y$. The first is bounded by a coefficient Hilbert norm. Testing it on FG and GF separately gives bounds for the other two.

Their squared dual coefficient norms have reciprocal weights: one on the constant coefficient and one half on the others. These are exactly the weights in the squared Fourier norms of the corresponding Hermitian diagonal compressions, after a common factor of four.

Section 3 completes the comparison. The numerical-range condition makes $R + R^*$ positive, so its compression to a two-by-two block bounds the off-diagonal boundary norm by the two diagonal norms. The cross term retains all coefficients of the first functional, including its constant coefficient. The resulting inequality gives the sharp bound two. Section 4 then shrinks analytic convex neighborhoods to an arbitrary numerical range and proves sharpness with a two-by-two nilpotent matrix. Appendix A supplies the exterior collar using classical conformal mapping and reflection theorems.

2 Exterior coordinates and coefficient estimates

In this section Ω is a bounded open convex set with regular real-analytic Jordan boundary. We work with a supplied exterior conformal map h that is one-to-one on $|\lambda| > r$, including infinity on the Riemann sphere, for some $0 < r < 1$. It maps $|\lambda| > 1$ onto $\mathbb{C} \setminus \overline{\Omega}$ and has a simple pole at infinity, with expansion

$$h(\lambda) = c\lambda + c_0 + O(\lambda^{-1}), \quad c \neq 0. \quad (2.1)$$

Appendix A supplies such a coordinate for every domain under consideration. Here we use it to compare polynomial coefficients with boundary norms.

The map h sends $\mathbb{T} = \{\lambda : |\lambda| = 1\}$ onto $\partial\Omega$. Put $q(\lambda) = \lambda h'(\lambda)$. On $\mathbb{T}$, the vector $q(\lambda)$ is an outward normal and $iq(\lambda)$ is a positively oriented tangent. Hence

$$\Re(\overline{q(\lambda)}(h(\lambda) - z)) \geq 0 \quad (\lambda \in \mathbb{T}, z \in \overline{\Omega}). \quad (2.2)$$

Figure 1 illustrates this supporting-line relation.

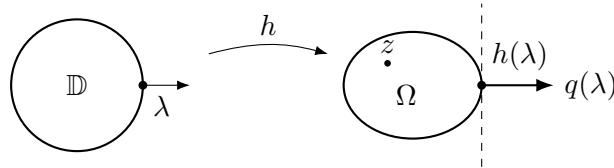


Figure 1: The map h sends the exterior of the circle to the exterior of the domain; the supporting line is shown schematically. Increasing the circle radius gives the outward normal $q(\lambda)$; convexity places every $z \in \overline{\Omega}$ on the inner side of the line.

Every circle integral below uses $d\tau(e^{it}) = dt/(2\pi)$.

We use the unnormalized Hilbert–Schmidt norm $|C|_2 = (\text{tr}(C^*C))^{1/2}$, which makes each finite-dimensional matrix space a Hilbert space.

Theorem 2.1 (Exterior Faber coefficients and the positive kernel). *Let Ω be a bounded convex domain with regular real-analytic Jordan boundary. Let h be meromorphic and one-to-one on $|\lambda| > r$ including infinity, where $0 < r < 1$, with a simple pole at infinity as in (2.1), and mapping $|\lambda| > 1$ onto $\mathbb{C} \setminus \overline{\Omega}$. Put $q(\lambda) = \lambda h'(\lambda)$ and define $b_k(z)$ by expansion at infinity:*

$$\frac{q(\lambda)}{h(\lambda) - z} = \sum_{k \geq 0} b_k(z) \lambda^{-k}. \quad (2.3)$$

Then $b_0 = 1$, and each b_k is a polynomial of degree k with leading coefficient c^{-k} . There are scalars s_{kj} , $k, j \geq 1$, such that

$$b_k(h(\mu)) = \mu^k + \sum_{j \geq 1} s_{kj} \mu^{-j} \quad (k \geq 1). \quad (2.4)$$

The function

$$s(\lambda, \mu) = \frac{q(\lambda)}{h(\lambda) - h(\mu)} - \frac{\lambda}{\lambda - \mu} = \sum_{k, j \geq 1} s_{kj} \lambda^{-k} \mu^{-j} \quad (2.5)$$

extends holomorphically across the diagonal and both infinities in $|\lambda|, |\mu| > r$. For every $r < \rho < 1$ there is a constant C_ρ with $|s_{kj}| \leq C_\rho \rho^{k+j}$. The series converges normally on the exterior product collar: it converges absolutely and uniformly on every compact subset, including subsets meeting either infinity. In particular it converges absolutely and uniformly on $\mathbb{T}^2$. The continuous kernel

$$K(\lambda, \mu) = 1 + s(\lambda, \mu) + \overline{s(\lambda, \mu)}$$

is nonnegative and has integral one in each variable with the other fixed.

Proof. Expanding the denominator in (2.3) after factoring $c\lambda$ shows that its k th coefficient is a polynomial of degree k with leading term $c^{-k}z^k$. The constant coefficient is one. In particular, the b_k form a basis of the scalar polynomials.

For the joint extension, consider

$$H(\lambda, \mu) = \frac{h(\lambda) - h(\mu)}{\lambda - \mu}.$$

Write $u = \lambda^{-1}$, $v = \mu^{-1}$ and $h(\lambda) = c\lambda + g(u)$, where g is holomorphic on $|u| < r^{-1}$. Then

$$H = c - uv \frac{g(u) - g(v)}{u - v}. \quad (2.6)$$

The divided difference of g is jointly holomorphic, with diagonal value $g'(u)$, so this formula also covers either or both infinities. Off the diagonal and away from the axes, H is nonzero by injectivity of h . On the diagonal it equals $h'(\lambda) \neq 0$; on either axis it equals c . Thus it has no zeros on the reciprocal bidisk.

Direct differentiation gives $s = \lambda \partial_\lambda H / H = -u \partial_u H / H$. This function is holomorphic and vanishes on both reciprocal axes. Its Taylor series has precisely the strictly mixed powers in (2.5). For any $r < \rho < 1$, Cauchy estimates on the closed reciprocal bidisk of radius ρ^{-1} give a constant C_ρ with $|s_{kj}| \leq C_\rho \rho^{k+j}$. On a compact subset of the exterior product collar, choose $\sigma > r$ below all finite coordinate moduli, and then $r < \rho < \min(1, \sigma)$. The resulting geometric double series dominates the Laurent series uniformly there; terms vanish on the reciprocal axes. This proves normal convergence throughout the product collar and absolute uniform convergence on the boundary. Comparing coefficients at $\lambda = \infty$ in (2.5), first for fixed finite μ in the exterior collar, yields (2.4).

For distinct $\lambda, \mu \in \mathbb{T}$, one has $2\Re(\lambda/(\lambda - \mu)) = 1$, and therefore

$$K(\lambda, \mu) = 2\Re \frac{q(\lambda)}{h(\lambda) - h(\mu)} \geq 0$$

by (2.2). Continuity gives the same sign on the diagonal. Every term of s and of its conjugate has nonzero frequency in each variable. Their integrals in either variable vanish, proving both normalizations. $\square$

To identify the basis, let $\Phi = h^{-1}$ in the exterior. Equation (2.4), which holds throughout the exterior collar, gives $b_k(z) - \Phi(z)^k = O(z^{-1})$ at infinity. Thus b_k is the polynomial part of Φ^k , the Faber polynomial associated with the chosen exterior coordinate. The usual normalization chooses the phase of h so that $c > 0$ [4, 5].

These mixed coefficients are differentiated Grunsky coefficients. The classical normalization gives a weighted coefficient inequality [6, equations (1.1)–(1.3)]; the unweighted estimate used here follows from convexity and the two angular marginal identities.

The two normalizations in Theorem 2.1 use the same exterior circle measure. Together they control the negative Fourier part of a boundary polynomial by its positive part, as follows.

Proposition 2.2 (The coefficient norm estimate). *For every positive integer m and every $M_m(\mathbb{C})$ -valued polynomial $G(z) = \sum_{k=0}^N b_k(z)C_k$,*

$$\sum_{k=0}^N |C_k|_2^2 \leq \int_{\mathbb{T}} |G(h(\mu))|_2^2 d\tau(\mu) \leq |C_0|_2^2 + 2 \sum_{k=1}^N |C_k|_2^2. \quad (2.7)$$

Proof. For $u \in L^2(\mathbb{T}; \mathcal{H})$ with values in a Hilbert space $\mathcal{H}$, define

$$(Tu)(\mu) = \int_{\mathbb{T}} K(\lambda, \mu) u(\lambda) d\tau(\lambda).$$

Positivity and the normalization in λ imply $\|(Tu)(\mu)\|^2 \leq \int K(\lambda, \mu) \|u(\lambda)\|^2 d\tau(\lambda)$. Integrating in μ and using the normalization in that variable shows that T is an L^2 contraction.

For $u(\lambda) = \sum_{k=1}^N C_k \lambda^k$, only the s term contributes to the integral, so

$$(Tu)(\mu) = \sum_{j \geq 1} \left(\sum_{k=1}^N s_{kj} C_k \right) \mu^{-j}. \quad (2.8)$$

All interchanges are justified by the absolute uniform convergence in Theorem 2.1. In particular,

$$\sum_{j \geq 1} \left| \sum_{k=1}^N s_{kj} C_k \right|_2^2 \leq \sum_{k=1}^N |C_k|_2^2.$$

By (2.4), the boundary value of G is the sum of its constant term C_0 , its positive part u , and its negative part Tu . These have disjoint Fourier supports. Parseval's identity gives

$$\int_{\mathbb{T}} |G \circ h|_2^2 d\tau = |C_0|_2^2 + \sum_{k=1}^N |C_k|_2^2 + \sum_{j \geq 1} \left| \sum_{k=1}^N s_{kj} C_k \right|_2^2,$$

which proves both inequalities. $\square$

For comparison with [16, Proposition 3.1], the companion's exterior map G and variables t, w correspond to h, λ, μ , respectively. The mixed correction is $s(\lambda, \mu) = B(\mu, \lambda)$, so $s_{kj} = \beta_{jk}$ in that paper's notation. In particular, (2.8) uses this coefficient orientation and does not require symmetry of the kernel.

3 Singular vectors and the amplified estimate

Fix $A \in M_n(\mathbb{C})$ with $W(A) \subset \Omega$. To estimate the norm of one polynomial evaluation, we form coefficient functionals from its left and right singular vectors. The resolvent supplies their coefficient arrays, while its positive Hermitian part compares the resulting boundary norms. This Hermitian field is the numerical-range double-layer kernel of [11, 10]; we prove the required positivity for the original matrix A .

Theorem 3.1 (The resolvent expansion and double-layer positivity). *Let Ω, h, q , and (b_k) satisfy the hypotheses and conclusions of Theorem 2.1. For any positive integer n and $A \in M_n(\mathbb{C})$ with $W(A) \subset \Omega$, on $\mathbb{T}$ let*

$$R(\lambda) = q(\lambda)(h(\lambda)I - A)^{-1}. \quad (3.1)$$

Then

$$R(\lambda) = \sum_{k \geq 0} b_k(A) \lambda^{-k}, \quad R(\lambda) + R(\lambda)^* \succeq 0.$$

For the fixed A, h , there are constants $C < \infty$ and $0 < \theta < 1$ such that $\|b_k(A)\| \leq C\theta^k$ for all $k \geq 0$. Thus the series converges absolutely and uniformly on $\mathbb{T}$. In particular, $\int_{\mathbb{T}} R d\tau = I$ and $\int_{\mathbb{T}} (R + R^*) d\tau = 2I$.

Proof. Every eigenvalue of A belongs to $W(A)$, by testing a unit eigenvector. Thus the exterior resolvent is defined on and outside the boundary. Compactness of the circle and strict spectral separation allow a smaller collar $|\lambda| > r_A$, with $r < r_A < 1$, on which it remains holomorphic, including infinity where $R(\infty) = I$. Expansion at infinity gives the coefficients $b_k(A)$: this follows either by the geometric resolvent series for large $|h(\lambda)|$ or by substituting A into the polynomial coefficient identities in (2.3). Choose $r_A < \theta < 1$. Cauchy estimates on the reciprocal circle of radius θ^{-1} give the stated geometric bound. They imply absolute uniform convergence on $\mathbb{T}$, and its constant coefficient $b_0(A) = I$ gives both integrals.

For $B = h(\lambda)I - A$, direct congruence gives

$$B^*(R + R^*)B = \overline{q(\lambda)}B + q(\lambda)B^*.$$

For each unit vector v , the quadratic form on the right is $2\Re(\overline{q(\lambda)}(h(\lambda) - v^*Av))$, which is nonnegative by (2.2). Invertibility of B proves positivity. $\square$

Theorem 3.2 (The bound on an analytic convex domain). *For positive integers n, m , let $\Omega \subset \mathbb{C}$ be bounded, open, and convex with regular real-analytic Jordan boundary, and let $A \in M_n(\mathbb{C})$ satisfy $W(A) \subset \Omega$. For every $M_m(\mathbb{C})$ -valued polynomial F with $\|F(z)\| \leq 1$ on $\partial\Omega$, one has $\|F[A]\| \leq 2$.*

Proof. Choose h from Lemma A.1 and apply Theorem 2.1 to this coordinate. Put $\gamma = \|F[A]\|$. There is nothing to prove if $\gamma = 0$. Otherwise choose unit singular vectors $x, y \in \mathbb{C}^n \otimes \mathbb{C}^m$ such that

$$F[A]x = \gamma y, \quad F[A]^*y = \gamma x. \quad (3.2)$$

Write $x = \sum_{i=1}^m X e_i \otimes e_i$ and $y = \sum_{i=1}^m Y e_i \otimes e_i$, where X, Y are n -by- m matrices and (e_i) is the standard basis of $\mathbb{C}^m$. For the resolvent in Theorem 3.1, define matrix functions

$$P = X^*RX, \quad Q = Y^*RY, \quad M = Y^*RX, \quad L = X^*RY.$$

Subscripts $k \geq 0$ denote the coefficients of λ^{-k} , so $M_k = Y^* b_k(A) X$, for example. By Theorem 3.1, all four series converge absolutely and uniformly on $\mathbb{T}$, and their coefficient sequences are square summable. Since $b_0 = 1$,

$$P_0 = X^* X = P_0^*, \quad Q_0 = Y^* Y = Q_0^*, \quad L_0^* = M_0. \quad (3.3)$$

Set $a^2 = \sum_{k \geq 0} |M_k|_2^2$. We will bound the squared boundary L^2 Hilbert–Schmidt norms of $P + P^*$ and $Q + Q^*$ above by $4a^2/\gamma^2$ each, and that of $M + L^*$ below by a^2 . Positivity will then compare these quantities and force $\gamma \leq 2$.

The off-diagonal coefficient functional. For a polynomial $G = \sum_k b_k C_k$, direct expansion in the tensor basis gives

$$y^* G[A] x = \sum_{k,i,j} (M_k)_{ij} (C_k)_{ij}. \quad (3.4)$$

Indeed, $C_k e_j = \sum_i (C_k)_{ij} e_i$ in the coefficient factor, leaving $(Y e_i)^* b_k(A) X e_j$ in the base factor. Equation (3.4) is the bilinear entry pairing of the two coefficient arrays. Cauchy–Schwarz gives

$$|y^* G[A] x| \leq a \left(\sum_k |C_k|_2^2 \right)^{1/2}. \quad (3.5)$$

Taking $G = F$ and using (3.2) shows that $a > 0$.

The two ordered tests. Polynomial evaluation respects multiplication in its given order. Explicitly, if $F(z) = \sum_i D_i z^i$ and $G(z) = \sum_j E_j z^j$, then

$$(FG)[A] = \sum_{i,j} A^{i+j} \otimes D_i E_j = F[A] G[A],$$

and the reversed product has coefficients $E_j D_i$. Hence

$$y^* (FG)[A] x = \gamma x^* G[A] x, \quad y^* (GF)[A] x = \gamma y^* G[A] y. \quad (3.6)$$

Write the full finite Faber expansions as

$$FG = \sum_k b_k U_k, \quad GF = \sum_k b_k V_k, \quad \|H\|_\partial^2 = \int_{\mathbb{T}} |H(h(\mu))|_2^2 d\tau(\mu).$$

The functional in (3.5) uses all M_k , so it applies to these full expansions for every finite degree. Apply it to the two identities in (3.6). The lower coefficient estimate in Proposition 2.2 and the two boundary multiplier bounds give

$$\begin{aligned} \gamma |x^* G[A] x| &\leq a \left(\sum_k |U_k|_2^2 \right)^{1/2} \leq a \|FG\|_\partial \leq a \|G\|_\partial, \\ \gamma |y^* G[A] y| &\leq a \left(\sum_k |V_k|_2^2 \right)^{1/2} \leq a \|GF\|_\partial \leq a \|G\|_\partial. \end{aligned}$$

Here $|F(z)G(z)|_2 \leq \|F(z)\| \|G(z)\|_2$ and $|G(z)F(z)|_2 \leq \|G(z)\|_2 \|F(z)\|$ are used in their respective orders. The upper coefficient estimate for G now gives

$$\max\{|x^* G[A] x|, |y^* G[A] y|\} \leq \frac{a}{\gamma} \left(|C_0|_2^2 + 2 \sum_{k \geq 1} |C_k|_2^2 \right)^{1/2}. \quad (3.7)$$

The entries paired with C_k in these two functionals are respectively those of P_k and Q_k , by the calculation in (3.4). The weighted coefficient norm on the right gives reciprocal weights on these two coefficient sequences:

$$|P_0|_2^2 + \frac{1}{2} \sum_{k \geq 1} |P_k|_2^2 \leq a^2/\gamma^2, \quad |Q_0|_2^2 + \frac{1}{2} \sum_{k \geq 1} |Q_k|_2^2 \leq a^2/\gamma^2. \quad (3.8)$$

For the first inequality, set $d_N = |P_0|_2^2 + \frac{1}{2} \sum_{k=1}^N |P_k|_2^2$ and test (3.7) with $C_0 = \overline{P_0}$, $C_k = \overline{P_k}/2$ for $1 \leq k \leq N$, and $C_k = 0$ otherwise; bars mean entrywise conjugation. The paired value is d_N , and the squared weighted test norm is also d_N . Hence $d_N \leq (a/\gamma)\sqrt{d_N}$, which gives $d_N \leq a^2/\gamma^2$ including when $d_N = 0$. Letting $N \rightarrow \infty$ proves the first bound. The same finite tests with Q_k prove the second.

The boundary norms of the three blocks. The compression of $R + R^*$ by $[X \ Y]$ has diagonal blocks $P + P^*$, $Q + Q^*$ and lower off-diagonal block $M + L^*$. We first compute their boundary norms. In the diagonal fields, (3.3) makes the constant term twice P_0 or Q_0 . For $k \geq 1$, the coefficients P_k and P_k^* occur at frequencies $-k$ and k and have equal Hilbert–Schmidt norms; the same holds for Q_k and Q_k^* . Parseval therefore gives four times the reciprocal weighted sum in (3.8):

$$\int_{\mathbb{T}} |P + P^*|_2^2 d\tau = 4 \left(|P_0|_2^2 + \frac{1}{2} \sum_{k \geq 1} |P_k|_2^2 \right) \leq 4a^2/\gamma^2, \quad (3.9)$$

and the same identity and bound hold for $Q + Q^*$.

For the cross field, $L_0^* = M_0$ makes the constant term $2M_0$. Its negative coefficients are M_k for $k \geq 1$ and its positive coefficients are L_k^* for $k \geq 1$. Consequently

$$\int_{\mathbb{T}} |M + L^*|_2^2 d\tau = 4|M_0|_2^2 + \sum_{k \geq 1} (|M_k|_2^2 + |L_k|_2^2) \geq a^2. \quad (3.10)$$

The positive block comparison. Theorem 3.1 gives $R + R^* \succeq 0$. Its compression by the n -by- $2m$ matrix $[X \ Y]$ is therefore

$$\Delta = \begin{pmatrix} P + P^* & L + M^* \\ M + L^* & Q + Q^* \end{pmatrix} \succeq 0.$$

For $J = \text{diag}(I_m, -I_m)$, the matrix $J\Delta J$ is also positive, so

$$0 \leq \text{tr}(\Delta^{1/2}(J\Delta J)\Delta^{1/2}) = \text{tr}(\Delta J\Delta J) = |P + P^*|_2^2 + |Q + Q^*|_2^2 - 2|M + L^*|_2^2.$$

Integrating this inequality and using (3.9) and (3.10) yields

$$2a^2 \leq 2 \int_{\mathbb{T}} |M + L^*|_2^2 d\tau \leq \int_{\mathbb{T}} (|P + P^*|_2^2 + |Q + Q^*|_2^2) d\tau \leq 8a^2/\gamma^2.$$

Since $a > 0$, this proves $\gamma \leq 2$. □

4 Enclosing domains and arbitrary numerical ranges

Theorem 3.2 applies on an analytic convex neighborhood. We now construct such neighborhoods even for compact convex sets with empty interior, and then shrink them to $W(A)$.

Lemma 4.1 (Analytic convex outer approximation). *Let $K \subset \mathbb{C}$ be nonempty, compact, and convex. There are bounded convex open domains Ω_j with regular real-analytic Jordan boundaries such that $K \subset \Omega_j$ and*

$$\sup_{z \in \overline{\Omega_j}} \text{dist}(z, K) \longrightarrow 0.$$

Proof. For $0 < \delta < 1$, choose a finite Euclidean δ -net D_δ of unit directions that includes $1, -1, i, -i$. Let $s_K(\nu) = \max_{z \in K} \Re(\bar{\nu}z)$ be the support function. Choose $t_\delta > 0$ with $|D_\delta|e^{-t_\delta\delta} < 1$, and define

$$\Phi_\delta(z) = \sum_{\nu \in D_\delta} \exp(t_\delta(\Re(\bar{\nu}z) - s_K(\nu) - \delta)), \quad \Omega_\delta = \{z : \Phi_\delta(z) < 1\}.$$

Each point of K satisfies $\Phi_\delta < 1$, so $K \subset \Omega_\delta$. If $\Phi_\delta(z) \leq 1$, each positive summand is at most one, and hence

$$\Re(\bar{\nu}z) \leq s_K(\nu) + \delta \quad (\nu \in D_\delta).$$

In particular, the four axis directions give

$$\begin{aligned} \min_{w \in K} \Re w - \delta &\leq \Re z \leq \max_{w \in K} \Re w + \delta, \\ \min_{w \in K} \Im w - \delta &\leq \Im z \leq \max_{w \in K} \Im w + \delta. \end{aligned}$$

Thus one fixed rectangle contains every closed sublevel for $0 < \delta < 1$.

The Hessian is a sum of positive multiples of $\nu\nu^\top$ after identifying $\mathbb{C}$ with $\mathbb{R}^2$. The four axis directions ensure it is positive definite. Thus Φ_δ is strictly convex and real analytic.

The gradient cannot vanish at level one: a critical point of a differentiable convex function is a global minimum, whereas every point of K has value less than one. The implicit function theorem therefore makes that level a regular real-analytic curve. The open sublevel set is bounded and convex with nonempty interior. Fix an interior point. Each ray from it crosses level one exactly once; convexity and boundedness give existence and uniqueness. Radial projection from the compact boundary to the circle is a continuous bijection, hence a homeomorphism. The boundary is a Jordan curve, and $\overline{\Omega}_\delta = \{\Phi_\delta \leq 1\}$.

It remains to bound the distance of this sublevel from K . For $z \in \overline{\Omega}_\delta \setminus K$, let $w \in K$ be nearest to z , put $d = |z - w|$ and $\nu = (z - w)/d$, and choose $\mu \in D_\delta$ with $|\mu - \nu| \leq \delta$. Choose $v \in K$ attaining $s_K(\mu)$. Since $w + t(v - w) \in K$ for $0 \leq t \leq 1$, minimality of w gives $\Re(\bar{\nu}(v - w)) \leq 0$. Using this inequality and the support inequality for μ , we obtain

$$\begin{aligned} (1 - \delta)d &\leq \Re(\bar{\mu}(z - w)) \\ &\leq \Re(\bar{\mu}(v - w)) + \delta \\ &\leq \delta|v - w| + \delta \leq \delta(1 + \text{diam } K). \end{aligned}$$

The first and third inequalities use $|\mu - \nu| \leq \delta$. Points of K have distance zero, so

$$\sup_{z \in \overline{\Omega}_\delta} \text{dist}(z, K) \leq \frac{\delta(1 + \text{diam } K)}{1 - \delta} \longrightarrow 0.$$

Taking, for example, $\delta_j = 1/(j + 1)$ completes the construction. $\square$

Proof of Theorem 1.1. Compactness of $W(A)$ follows from compactness of the unit sphere. We recall a short proof of the Toeplitz–Hausdorff convexity theorem [18, 12]. To join two numerical-range values, compress A to the span of their unit testing vectors. If that span has dimension

one there is nothing to prove. In dimension two, its quadratic form on unit vectors (u, v) is a complex-valued real-affine function of

$$(2\Re(\bar{u}v), 2\Im(\bar{u}v), |u|^2 - |v|^2),$$

which ranges over the full unit sphere in $\mathbb{R}^3$. The linear part of a map from $\mathbb{R}^3$ to $\mathbb{R}^2$ has a nonzero kernel. Any point of the unit ball can be moved in a kernel direction to the sphere without changing its image. Thus the images of the sphere and ball coincide, and this image is convex. Its segment between the chosen values belongs to the numerical range of the compression and hence to $W(A)$.

Apply Lemma 4.1 with $K = W(A)$. By scaling Theorem 3.2,

$$\|P[A]\| \leq 2 \max_{z \in \bar{\Omega}_j} \|P(z)\|.$$

If this maximum is zero, each entry of P vanishes on an open set and the polynomial is identically zero, making the inequality immediate. Otherwise divide P by the maximum to invoke that theorem. All $\bar{\Omega}_j$ lie in one compact set. Uniform continuity of the fixed function $z \mapsto \|P(z)\|$, the containment of K , and the uniform distance convergence imply convergence of these maxima to $\max_K \|P\|$. This proves (1.1) without any interior assumption on K .

For sharpness take $n = 2$, $m = 1$, $P(z) = z$, and

$$A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

Its norm is two, while $|x^*Ax| = 2|x_1x_2| \leq 1$ for every unit x , with equality at $x_1 = x_2 = 1/\sqrt{2}$. Thus the right-hand maximum in (1.1) is one. No smaller universal constant is possible. $\square$

A The exterior conformal collar

We record the classical conformal inputs and prove the continuation and global injectivity required in Section 2.

Lemma A.1 (Exterior coordinate and a univalent collar). *Let $\Omega \subset \mathbb{C}$ be a bounded convex domain whose boundary is a regular real-analytic Jordan curve. There are $0 < r_0 < 1$ and a map h , meromorphic on $\{\lambda : |\lambda| > r_0\} \cup \{\infty\}$, which is one-to-one there as a map of the sphere, has a simple pole at infinity, and maps $|\lambda| > 1$ conformally onto $\mathbb{C} \setminus \bar{\Omega}$. It maps the unit circle onto $\partial\Omega$ and has an expansion*

$$h(\lambda) = c\lambda + c_0 + \sum_{j \geq 1} c_j \lambda^{-j}, \quad c \neq 0.$$

For λ on the unit circle, $q(\lambda) = \lambda h'(\lambda)$ is a nonzero outward normal vector at $h(\lambda)$. In particular,

$$\operatorname{Re}(\overline{q(\lambda)}(h(\lambda) - z)) \geq 0 \quad (z \in \bar{\Omega}).$$

Proof. Fix $z_* \in \Omega$ and apply $z \mapsto 1/(z - z_*)$, with infinity mapped to zero, to the spherical exterior $(\mathbb{C} \setminus \bar{\Omega}) \cup \{\infty\}$. Its image U is a bounded simply connected domain containing zero, with a regular real-analytic Jordan boundary. Indeed, each ray from z_* meets $\partial\Omega$ at a unique positive radius; inversion turns the exterior portions of these rays into a star-shaped domain about zero. The Riemann mapping theorem [2, Chapter 6, Section 1.1, Theorem 1] supplies a conformal bijection

$g : \mathbb{D} \rightarrow U$ with $g(0) = 0$. Carathéodory's Jordan-domain theorem [2, Chapter 6, Section 1.2] extends g to a homeomorphism $\overline{\mathbb{D}} \rightarrow \overline{U}$.

We justify the stronger boundary continuation needed below. A regular real-analytic boundary arc has a real-analytic parametrization with nonzero derivative. Its holomorphic continuation and the holomorphic inverse-function theorem provide a local coordinate in which the arc is a real interval and the domain is one side of that interval. Choose such coordinates at $\zeta \in \partial\mathbb{D}$ and $p = g(\zeta) \in \partial U$. Continuity of the boundary extension lets us shrink the source neighborhood so that g maps its intersection with $\overline{\mathbb{D}}$ into the target chart. The composition representing g in these coordinates is holomorphic on a half disk, continuous on its real interval, and real-valued there. Schwarz reflection extends it holomorphically across the interval [2, Chapter 4, Section 6.5; Chapter 6, Sections 1.3–1.4]. Apply the same procedure to g^{-1} . Shrink the neighborhoods so that the reflected maps can be composed. Their compositions equal the identity on the original domain side, and hence throughout a smaller neighborhood by the identity theorem. Thus the continued g has nonzero derivative at each boundary point.

To patch the local continuations, choose around every $\zeta \in \partial\mathbb{D}$ a disk $B(\zeta, R_\zeta)$ carrying such a continuation, agreeing with g on its intersection with $\mathbb{D}$. Select finitely many of the third-radius disks $V_i = B(\zeta_i, R_i/3)$ that cover the circle, and denote the larger disks by $U_i = B(\zeta_i, R_i)$. If $V_i \cap V_j \neq \emptyset$ and $R_i \geq R_j$, then $|\zeta_i - \zeta_j| < (R_i + R_j)/3 \leq 2R_i/3 < R_i$. The connected lens $U_i \cap U_j$ therefore contains ζ_j and meets $\mathbb{D}$. The two continuations coincide there by the identity theorem. They consequently patch with g to a holomorphic map on $\mathbb{D} \cup \bigcup_i V_i$, a neighborhood of the closed disk.

The extension is injective on a smaller neighborhood of the closed disk. Otherwise, distinct colliding pairs arbitrarily close to that compact set would have subsequential limits in the closed disk. Continuity and injectivity on the closed disk identify their limits. Both points of each pair would then eventually lie in one local inverse neighborhood of that common limit, a contradiction. By compactness, this injective neighborhood contains the disk $\{w : |w| < R_0\}$ for some $R_0 > 1$. Thus g is univalent on that disk. Its sole zero there is 0, and that zero is simple. Set

$$h(\lambda) = z_* + \frac{1}{g(1/\lambda)}, \quad r_0 = R_0^{-1}.$$

This gives all mapping and extension assertions, including the simple pole with coefficient $c = 1/g'(0) \neq 0$.

Finally, at $\lambda = e^{i\theta}$ the radial and angular derivatives are $q(\lambda)$ and $iq(\lambda)$, respectively. They are perpendicular and nonzero. Increasing the radial parameter enters the exterior of $\overline{\Omega}$, so $q(\lambda)$ points outward. The supporting half-plane property of a smooth convex boundary proves the final inequality. $\square$

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